



QuantVLA: Scale-Calibrated Post-Training Quantization for Vision-Language-Action Models

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Abstract

Vision-language-action (VLA) models unify perception, language, and control for embodied agents but face significant challenges in practical deployment due to rapidly increasing compute and memory demands, especially as models scale to longer horizons and larger backbones. To address these bottlenecks, we introduce **QuantVLA**, a training-free post-training quantization (PTQ) framework that, to our knowledge, is the first PTQ approach for VLA systems and the first to successfully quantize a diffusion transformer (DiT) action head. **QuantVLA** incorporates three scale-calibrated components: (1) a selective quantization layout that integerizes all linear layers in both the language backbone and the DiT while keeping attention projections in floating point to preserve the original operator schedule; (2) attention temperature matching, a lightweight per-head scaling mechanism that stabilizes attention logits and is folded into the dequantization scales at inference; and (3) output head balancing, a per-layer residual interface calibration that mitigates post-projection energy drift. The framework requires no additional training, uses only a small unlabeled calibration buffer, and supports integer kernels for low-bit weights and activations while leaving the architecture unchanged. Across representative VLA models on LIBERO, **QuantVLA exceeds** the task success rates of full-precision baselines, achieves about **70% relative memory savings** on the quantized components, providing a practical pathway toward scalable low-bit embodied intelligence under strict compute, memory, and power constraints.

1. Introduction

Vision-Language-Action (VLA) models [52] represent an important step toward embodied multimodal intelligence. They allow robots to parse visual observations together with natural language instructions and to output executable actions. Recent progress in large pretrained language models (LLMs) [35, 41], vision-language models (VLMs) [3, 14], and Diffusion Transformer (DiT) architecture [21, 27] has turned VLA systems into a unified interface that connects perception, high-level reasoning, and low-level control. Building on these backbones, $\pi 0.5$ [11], OpenVLA [13], and GR00T N1.5 [4] strongly unify vision, language, and action, achieving strong performance in robotic manipulation and reasoning. As embodied models scale toward foundation size, improving efficiency is critical for deployment on robotic platforms with limited compute and memory under strict constraints.

However, large-scale VLA architectures incur substantial cost, primarily from downstream reasoning and control rather than visual perception [44], where high-dimensional hidden states and sequential decoding dominate compute and memory, limiting deployment in embedded and mobile settings.

Existing work on improving the efficiency of VLA systems can be roughly grouped into two families: methods that design more efficient VLA models [32, 37] and methods that build efficiency frameworks around existing policies [40, 44]. However, these optimizations act primarily on the vision encoder and do not directly address the efficiency or robustness of the language backbone and the diffusion-based policy head [7, 23]. In practice, the DiT action head is a major contributor to computation and memory and is tightly coupled to the language backbone, so its behavior strongly affects performance. Yet existing efficiency frameworks typically leave this component unchanged, in part because it is tightly coupled to upstream reasoning and diffi-

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Figure 1. **Comparison of representative VLA efficiency frameworks.** (1) TinyVLA focuses on compact multimodal transformers and lightweight diffusion-policy heads for architectural efficiency; (2) EfficientVLA accelerates inference by pruning redundant language layers and reusing intermediate representations; (3) VLA-Cache improves throughput through key-value reuse and static caching of vision tokens; (4) MoLe-VLA adopts mixture-of-layers routing to dynamically skip computation in the language module; and (5) **QuantVLA** introduces a training-free PTQ framework that low-bit quantizes both language and action modules without altering the model architecture.

cult to modify without degrading stability, and instead focus on the visual front end, which means that the main opportunities for reducing the cost of reasoning and action generation remain underexploited. Besides, post-training quantization (PTQ) [24] methods such as SmoothQuant [39] and DuQuant [17] demonstrate that careful precision allocation can reduce memory use and improve efficiency, but they are primarily developed for large language or vision language models and do not capture the heterogeneous activation and precision behaviors of downstream reasoning and action modules in VLA systems. To provide a clearer overview of these acceleration paradigms, Fig. 1 summarizes representative VLA frameworks across language, action, and vision components.

As shown in Fig. 1, most existing efficient VLA models or methods either redesign transformer and diffusion blocks or add routing and caching around an unchanged policy, while almost none operate directly on the DiT action head or treat precision allocation as a primary design choice. In particular, current frameworks keep the policy head in full precision and focus on the visual front end. This leaves an important opportunity unused: if one can apply PTQ to the highly sensitive DiT head without degrading performance, PTQ becomes a powerful tool for VLA models, since it can substantially reduce memory and bandwidth without retraining, which is especially valuable for large VLA policies that couple a language backbone with a diffusion policy head.

Therefore, to address these gaps, we introduce **QuantVLA**, a scale-calibrated PTQ framework specifically designed for VLA models. We first analyze why the DiT-based action head is fragile under upstream quantization, showing that quantization-induced scale drift changes the effective logits temperature and the residual stream energy, which explains its strong sensitivity to activation changes and precision loss. Guided by this analysis, QuantVLA performs selective post-training quantization over the language and action pathways and introduces two lightweight calibration mechanisms that restore the key scales after quantization. The resulting design led QuantVLA to achieve about

70% relative memory savings on the quantized modules, while even exceeding the LIBERO [22] task success rates of the full precision baseline as shown in the middle and right panel of Fig. 1. In conclusion, our main contributions are summarized as follows:

1. We provide the first systematic analysis of quantization sensitivity in VLA models with DiT action heads, identifying key failure modes that explain the breakdown of PTQ.
2. We propose **QuantVLA**, the first rotation-based, training-free PTQ framework for VLA models, achieving state-of-the-art performance under low-precision inference while enabling substantial memory savings under low-precision deployment.

2. Related Work

2.1. Vision-Language-Action Models

VLA models unify perception, reasoning, and control within a single multimodal policy. Existing systems can be grouped into several categories described below. Encoder-decoder approaches such as ALOHA and ACT [50], RT-1 [6], and HPT [36] train Transformer networks from scratch to map visual observations and robot states to actions, achieving high accuracy but limited generalization. Pretrained language or vision language models such as RT-2 [54] and OpenVLA [13] represent actions as autoregressive tokens, which enables open vocabulary reasoning but weakens temporal smoothness. Diffusion-based policies such as Diffusion Policy [7] and RDT-1B [23] generate continuous action trajectories via DiT architectures.

Besides, hybrid architectures that combine language reasoning and diffusion-based control have recently become dominant [12]. OPENPI π_0 [5] and OPENPI $\pi_{0.5}$ [11] unify vision and language inputs within a single diffusion transformer DiT [27], which tightly couples semantic reasoning and low level actuation, while GR00T N1.5 [4] extends this paradigm through a dual system design where a vision-language interpreter grounds semantics and a DiT trained with flow matching [21] objectives generates pre-

cise humanoid motion. As VLA models scale in horizon and size, deployment is increasingly constrained by the reasoning and action stack, where the language backbone and diffusion-based policy head dominate compute and memory.

2.2. Efficient and Compact VLA Models

Recent work has explored efficient and compact vision-language-action (VLA) models that reduce deployment cost by designing lightweight architectures, smaller backbones, or specialized inference pipelines. TinyVLA [37] builds compact multimodal transformers with lightweight diffusion policy heads to achieve faster inference and improved data efficiency. SmolVLA [32] targets affordable robotics by adopting a small VLA architecture together with an asynchronous inference stack to keep control latency low. FLOWER [29] and X-VLA [51] further explore architectural simplification and alternative action formulations to improve efficiency at reduced model scales.

These approaches improve efficiency via new model designs and training pipelines. In contrast, QuantVLA is a post-training quantization framework that preserves the original architecture and training, remaining orthogonal to compact VLA design and applicable to both large foundation VLAs and smaller efficient variants.

2.3. Efficiency Frameworks for Pretrained VLAs

Another line of work improves the efficiency of pretrained VLA models by optimizing the inference framework without redesigning the underlying policy. EfficientVLA [44] accelerates inference by pruning redundant language layers, selecting compact visual tokens, and reusing intermediate representations. VLA-Cache [40] reduces computational overhead by detecting unchanged visual observations across frames and reusing cached key-value features during rollouts. MoLe-VLA [47] further introduces mixture-of-layers routing to dynamically skip non-essential computation in the language backbone.

QuantVLA instead targets numerical precision and post-training deployment efficiency, quantizing both the language backbone and diffusion-based action head without altering execution order or adding routing logic. Besides, recent work also explores efficient tokenization and action discretization for VLAs, such as FAST [28], BEAST [53], and OmniSAT [2], which reduce sequence length or improve token utilization. These approaches operate at the representation level and are complementary to QuantVLA, which focuses on numerical precision and post-training deployment efficiency.

2.4. Post-Training Quantization

Post-training quantization (PTQ) has been extensively studied as an effective approach to reduce memory usage

for pre-trained models [18, 42, 43]. The basic RTN quantization formulation is summarized in Appendix A, from Eq. 18 to Eq. 21. PTQ methods can be broadly categorized into weight-only quantization [8, 9, 20, 45] and weight-activation quantization [19, 30, 38, 49]. Our work focuses on the latter, specifically on exploring ultra low-bit weight-activation quantization. For large language models (LLMs), SmoothQuant [39] rescales activation and weight channels to suppress outliers and stabilize low-bit transformer inference. Rotation-based approaches [1, 10, 17, 34] further apply orthogonal transforms to spread activation outliers across channels. DuQuant [17] applies dual-path transformations that combine block-orthogonal rotations with per-channel smoothing, effectively redistributing outliers and improving robustness under low-bit precision. For diffusion transformers (DiTs), SVDQuant [15] protects activation outliers by introducing low-rank residual branches, while ViDiT-Q [48] employs fine-grained grouping and dynamic quantization to better adapt to activation statistics. However, directly applying them to VLA models is challenging, as VLA pipelines tightly couple multimodal reasoning with diffusion-based action generation in a single policy. Scale drift across modalities and along the diffusion rollout violates the assumptions underlying existing PTQ techniques. In particular, quantization-induced scale mismatch can distort the effective attention-logits temperature and the residual-stream energy in diffusion-policy heads, which makes stable low-bit control substantially harder than unimodal transformers. A key open problem is how to design quantization schemes that remain stable under such tight multimodal-diffusion coupling while still achieving low-bit efficiency for VLA control.

3. Method

3.1. Preliminaries on Diffusion-based VLA Models

We study Vision-Language-Action (VLA) systems whose action head is a Diffusion Transformer (DiT) [27, 31]. At each control step, a short history of RGB frames is embedded via a pretrained vision encoder, such as SigLIP2 [46] or DINOv2 [26], to produce image tokens. Concurrently, the natural language instruction is tokenized and embedded by a pretrained language backbone. The visual and textual tokens are projected into a shared transformer space, where attention merges perception with the instruction context to form a task-conditioned representation F_{VL} .

The policy head, a Diffusion Transformer, is conditioned on F_{VL} , on robot proprioception, and on a diffusion timestep t . It iteratively updates an action latent according to:

$$x_{t-1} = f_{\theta}(x_t, F_{VL}, t). \quad (1)$$

After T refinement steps, the final latent x_0 is decoded into

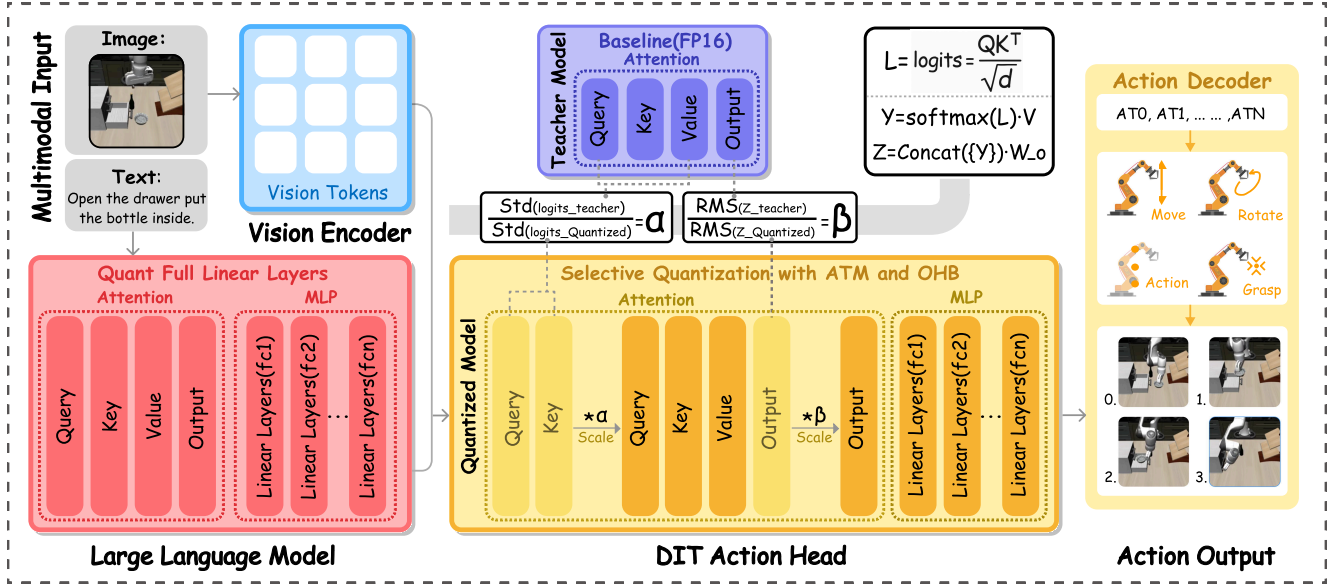


Figure 2. Overview of **QuantVLA** for VLAs with a DiT-based action head. The framework is training-free and preserves the original architecture and operator schedule. It combines: (1) a selective quantization layout that integerizes all linear layers in the LLM and all MLP layers in the DiT while keeping the attention projections Q, K, V, O in floating point; (2) *Attention Temperature Matching* (ATM), a per-head scalar α that aligns teacher–student logits and is folded into dequantization scales; and (3) *Output Head Balancing* (OHB), a per-layer scalar β that matches post-projection energy at the residual interface.

the action. For tokenized policies, the output is a sequence of discrete action tokens. In this formulation, the diffusion transformer denotes the architecture of the policy head. Flow matching [21] denotes the learning objective used to fit f_θ and views the same iterative refinement as a conditional ordinary differential equation (ODE) [33] that trains the network to predict a velocity field transporting x_t toward executable actions.

3.2. Post-Training Quantization Setup and Emergent DiT Sensitivity

3.2.1. DuQuant Reparameterization.

Among PTQ variants, DuQuant [17] is empirically the most stable under aggressive bit widths for transformer stacks. It does so via an invertible reparameterization of each linear layer that (i) applies per-channel smoothing with a diagonal matrix Λ , (ii) performs block-orthogonal rotations $\hat{R}_{(1)}, \hat{R}_{(2)}$, and (iii) inserts a zigzag channel permutation to redistribute outliers, which preserves the original linear map and makes activations and weights more amenable to low-bit quantization. Inspired by these robustness and outlier-redistribution properties, we adopt a similar reparameterization for linear layers in VLA models to improve stability under quantization. Additional implementation details are provided in the Appendix B.

3.2.2. Challenges in Implementing Quantization for VLA

While the reparameterization in the previous subsection improves low-bit robustness at the layer level, deploying it to tightly coupled VLM stacks exposes two issues. First, quantizing the upstream language backbone perturbs intermediate representations that condition the DiT action policy, and the perturbation propagates downstream as input drift. Second, the DiT head must emit precise action tokens for real robots, which means small rounding and scale mismatches will translate into control errors.

As clarified in Appendix C, dequantization scales control two deterministic factors in DiT, which are the effective logits temperature through $s_q s_k$ and the residual stream energy through $s_v s_o$. It explains how logits and energy transfer from the language backbone to DiT when no perturbation is present. Building on this baseline, we now analyze how quantization errors propagate and accumulate.

Building on the deterministic transfer in Appendix C, we now develop a first-order analysis of error propagation to show that the distribution reaching attention is perturbed when DuQuant is applied to the upstream language backbone and the DiT feed-forward blocks. Let X_T denote the teacher input and let $X_Q = X_T + \varepsilon_{\text{up}}$ denote the input under quantization. Even if the attention weights remain in floating point, this perturbation propagates linearly.

$$\begin{aligned} Q_Q &= X_Q W_q = Q_T + \varepsilon_{\text{up}} W_q, \\ K_Q &= X_Q W_k = K_T + \varepsilon_{\text{up}} W_k. \end{aligned} \quad (2)$$

Define the pre-softmax logits L :

$$L_T = \frac{Q_T K_T^\top}{\sqrt{d}}, \quad L_Q = \frac{Q_Q K_Q^\top}{\sqrt{d}}, \quad (3)$$

and let $\Delta L = L_Q - L_T$. Keeping only first-order terms yields

$$\Delta L \approx \frac{1}{\sqrt{d}} \left((\varepsilon_{\text{up}} W_q) K_T^\top + Q_T (\varepsilon_{\text{up}} W_k)^\top \right) + \Delta L_{\text{local}}, \quad (4)$$

where ΔL_{local} aggregates local rounding and scale mismatch from the quantized activations that feed Q and K from the output projection. Let $A = \text{softmax}(L)$ and let $J_{\text{softmax}}(\cdot)$ denote its Jacobian. The attention update satisfies

$$A_Q \approx A_T + J_{\text{softmax}}(L_T) \Delta L. \quad (5)$$

Now include the output head. Write the value path and the quantized output projection as

$$V_Q = X_Q W_v = V_T + \varepsilon_{\text{up}} W_v. \quad (6)$$

The teacher and quantized outputs are

$$O_T = A_T V_T W_{o,T}, \quad O_Q = A_Q V_Q W_{o,Q}. \quad (7)$$

A first-order expansion around the teacher gives

$$\begin{aligned} \Delta O &\approx J_{\text{softmax}}(L_T) \Delta L V_T W_{o,T} \\ &\quad + A_T \varepsilon_{\text{up}} W_v W_{o,T} \\ &\quad + A_T V_T \delta W_o \\ &\quad + \Delta O_{\text{local}}. \end{aligned} \quad (8)$$

According to Eq. 8, quantization in DiT introduces two systematic drifts. First, variance changes in Q and K alter the scale of attention logits, which shifts the effective temperature of the softmax and moves attention entropy away from the teacher distribution. This temperature bias does not vanish within a single layer. It is carried forward by the attention outputs and persists across layers. Second, after multiple head concatenation and the output projection, the amplitude of the attention output exhibits a systematic change. This modifies the residual injection gain and the operating point of layer normalization. In deep DiT stacks, these two drifts accumulate through residual connections and normalization, which degrades stability and overall performance.

3.3. QuantVLA Framework

In this section, we present **QuantVLA**, a training-free and deployment-oriented framework that preserves the original model architecture and operator schedule while addressing the two dominant sensitivity factors identified in Sec. 3.2.2. As shown in Fig. 2, QuantVLA combines selective quantization with two lightweight calibrations. As

noted above, quantizing every linear layer in the LLM and the DiT head causes errors to accumulate along the attention and residual pathways. Guided by this analysis, we integerize all linear layers in the LLM and adopt a selective DiT quantization layout, while keeping the attention projections W_q , W_k , W_v , and W_o in floating point to avoid amplifying the two drifts identified in Sec. 3.2.2 since they are most sensitive to upstream distribution shifts and directly determine the stability of the softmax distribution and the residual injection. This layout mitigates the dominant sources of drift in DiT under low bit widths. However, integerizing the upstream LLM can still bias the statistics that reach the DiT head. To compensate for this cross-module drift in VLA pipelines, we introduce two lightweight calibrations, **Attention Temperature Matching (ATM)** and **Output Head Balancing (OHB)**, as shown in Fig. 2. Both are estimated from an unlabeled calibration buffer and folded into the de-quantization scales, so the operator schedule and integer GEMMs remain unchanged. In QuantVLA, ATM and OHB are instantiated specifically at the language-to-action interface of VLA pipelines, where quantized language features condition the DiT head and induce the strongest scale drift. ATM uses per-head temperature scalars to align the logits distribution through Q and K , preventing attention from becoming overly sharp or overly flat under upstream VLA quantization. OHB uses per-layer output scalars to align the post-projection energy through W_o , restoring the residual injection gain and the operating point of layer normalization in the DiT head.

Specifically, we calibrate ATM by matching the dispersion of the teacher and quantized logits L defined in Eq. 3. We estimate a scalar α from a small unlabeled calibration buffer and apply it at inference time as follows

$$\alpha_{\text{raw}} = \frac{\text{Std}(L_T)}{\text{Std}(L_Q) + 10^{-6}}. \quad (9)$$

We then confine the correction to a safe range to avoid over-cooling or over-heating the attention distribution

$$\alpha = \text{clip}(\alpha_{\text{raw}}, \alpha_{\text{min}}, \alpha_{\text{max}}). \quad (10)$$

Next, we apply a neutrality band ε to ignore negligible differences and reduce sensitivity to calibration noise

$$\text{if } |\log \alpha| < \varepsilon \text{ then } \alpha = 1. \quad (11)$$

Therefore, the quantized logits become

$$L_Q = \frac{L_T}{\alpha}. \quad (12)$$

We next match the post-projection energy at the residual interface to stabilize the residual injection gain and the operating point of layer normalization. The activation of the output head at the layer l is

$$Z_l = \text{Concat}\{A_{l,h} V_{l,h}\} W_{o,l} + b_{o,l}. \quad (13)$$

We measure per-layer energy using RMS for the teacher and the quantized, and directly form a teacher-to-student ratio

$$\beta_{\text{raw}}(l) = \frac{\text{RMS}(Z_{T,l})}{\text{RMS}(Z_{Q,l}) + 10^{-6}} \quad (14)$$

Similarly to ATM, we confine this factor to a safe range and apply a neutrality band

$$\beta(l) = \text{clip}(\beta_{\text{raw}}(l), \beta_{\text{min}}, \beta_{\text{max}}), \quad (15)$$

$$\text{if } |\log \beta(l)| < \varepsilon \text{ then } \beta(l) = 1. \quad (16)$$

Finally, we rescale the activation of the output head that enters the residual path

$$Z_Q = \frac{Z_l}{\beta(l)}. \quad (17)$$

Building on these techniques, QuantVLA combines a flexible selection of quantized linear layers to counter the sensitivities identified in Sec. 3.2.2. By integerizing all linear layers in the LLM and adopting a selective quantization layout in the DiT while keeping the attention projections in floating point, we avoid compounding errors at the most fragile interfaces. Furthermore, ATM aligns the teacher and student logits statistics to correct attention temperature drift, whereas OHB restores the residual injection gain by matching the output head energy. Crucially, ATM and OHB are realized as tiny per-head and per-layer scalars that are estimated once from an unlabeled calibration buffer and folded into existing dequantization scales. They introduce no new operators or activations, require no additional buffers, preserve the original operator schedule and integer GEMMs, and therefore incur no additional GEMM computation during inference. The only overhead is scalar folding performed once during calibration. Based on these steps, we stabilize the DiT action head under low bit widths without retraining.

4. Experiment

4.1. Experimental Settings

Model and Benchmark. We evaluate on two state-of-the-art VLA policies, OpenPI $\pi 0.5$ [11] and GR00T N1.5 [4], both employing a DiT-based action head that maps fused visual–language features to action sequences. The models span complementary regimes, where $\pi 0.5$ prioritizes efficient inference and GR00T N1.5 offers higher capacity and richer action modeling, and this breadth enables a robust assessment across different coupling strengths between perception and control. Evaluation uses the LIBERO [22] simulator with four task suites that target distinct capabilities: Spatial tests relational reasoning and precise placement, Object focuses on object-centric grasping and manipulation, Goal measures instruction-to-goal alignment and

condition satisfaction, and Long examines temporal decomposition and control of accumulated error. We report the success rate under the standard LIBERO protocol to ensure fair comparison and reproducibility.

Implementation Details. We adopt our method with a W4A8 setting. Scales are estimated from a small unlabeled calibration buffer and folded into dequantization at inference. For stability matching, the α of ATM and β of OHB are clipped to a safe range of ± 0.4 before being folded into the scales and using a neutrality band ε of 0.03. All experiments are conducted on NVIDIA A100 GPUs. More details are shown in Appendix D

4.2. Empirical Validation of the Selective Quantization Layout

As established in Sec. 3.2.2, quantization errors introduced by the upstream language backbone perturb the attention temperature and the residual energy in the DiT action head, which renders the attention projections and the residual interface particularly sensitive. To limit this cross-module drift, we compare several layer selection schemes that quantize the LLM only, the action head only, both modules in full, or the LLM together with the DiT MLP. We evaluate these alternatives on OpenPI $\pi 0.5$ and GR00T N1.5 within LIBERO, and we isolate the effect of layer choice by disabling ATM and OHB in this ablation so that we observe the pure quantization outcome. The results in Table 1 show a consistent pattern across models and suites, as quantizing the entire action head or the full stack leads to the largest degradation, most notably in the long-horizon task, whereas quantizing the LLM together with the DiT MLP remains closest to the baseline while retaining the memory benefits of integer computation counted over the LLM and DiT components, which aligns with our theoretical analysis in Sec. 3.2.2. Consequently, we fix the layer selection to all linear layers in the LLM and the MLP blocks in the DiT while leaving Q , K , V , and O in floating point for all subsequent experiments.

4.3. Effect of ATM and OHB Calibration

In this section, we empirically verify that ATM and OHB restore logits statistics and output energy. Fig. 3 evaluates three configurations on GR00T N1.5: the floating-point teacher, QuantVLA without ATM and OHB calibration, and QuantVLA with ATM in the left panel or with OHB in the right panel. In the left panel, ATM reduces the mismatch in logits Std and moves each attention block toward the teacher, which shows that temperature shifts caused by quantization are corrected. In the right panel, OHB aligns the attention output RMS after the output projection with the teacher, which mitigates residual-stream energy drift and stabilizes the downstream residual path. Across blocks,

Model	Precision	Layer Selection	Layer Nums	LIBERO					Memory (GB) (LLM+DiT)
				Spatial	Object	Goal	Long	Avg.	
π 0.5	FP16	No Quantization	0	98.5%	99.0%	97.5%	93.5%	97.1%	4.27
	W4A8	LLM	126	98.0%	98.5%	97.5%	92.0%	96.5%	1.58
	W4A8	DiT	126	81.5%	94.5%	71.5%	39.0%	71.6%	3.85
	W4A8	LLM+DiT	252	86.0%	97.5%	71.5%	50.0%	76.3%	1.17
	W4A8	LLM+DiT (MLP)	180	98.0%	97.0%	94.5%	92.0%	95.4%	1.28
GR00T N1.5	FP16	No Quantization	0	92.0%	92.0%	86.0%	76.0%	86.5%	2.02
	W4A8	LLM	84	86.0%	92.0%	80.0%	80.0%	84.5%	1.25
	W4A8	DiT	96	88.0%	80.0%	86.0%	78.0%	83.0%	1.49
	W4A8	LLM+DiT	180	66.0%	70.0%	68.0%	76.0%	70.0%	0.74
	W4A8	LLM+DiT (MLP)	116	90.0%	86.0%	80.0%	74.0%	82.5%	0.91

Table 1. Selective layer-quantization results under the QuantVLA architecture without ATM/OHB calibration for π 0.5 and GR00T N1.5 on LIBERO.

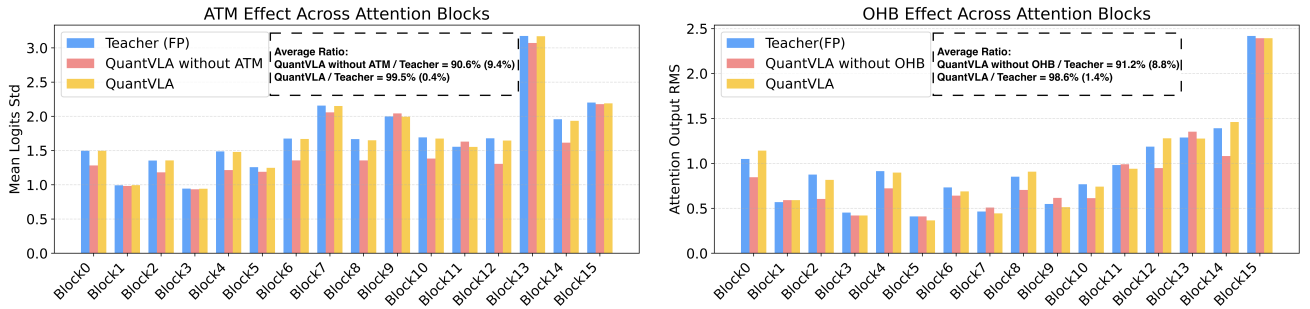


Figure 3. ATM and OHB effects across attention blocks. **(Left)** shows logits standard deviation. **(Right)** shows attention output RMS after the output projection. The figure reports three configurations: the teacher model in floating point without quantization, the quantized baseline with LLM and DiT MLP integerized, and QuantVLA with ATM in the left panel or QuantVLA with OHB in the right panel, which are evaluated on the GR00T N1.5 model.

the calibrated curves consistently narrow the gap to the teacher, especially in deeper layers, confirming that ATM corrects logits statistics and OHB corrects output energy. We therefore include both components in all subsequent experiments.

4.4. QuantVLA Results in LIBERO Simulation

Main Results on LIBERO. Table 2 reports the performance of different quantization techniques on OpenPI π 0.5 and GR00T N1.5 within the LIBERO simulator. We compare two representative approaches: DuQuant and the proposed QuantVLA framework. DuQuant can be successfully applied to both the LLM and the DiT, but its task accuracy drops significantly under this configuration, for example, on π 0.5 the average success rate falls to 76.3%, and on GR00T N1.5 it reaches 70%. These outcomes suggest that methods designed for unimodal or loosely coupled settings do not transfer to highly coupled VLA systems. In contrast, QuantVLA is the first framework to achieve effective PTQ on VLA models. By combining selective layer quantization with ATM and OHB calibration, QuantVLA

not only maintains stable performance but also surpasses the baseline on several task suites. On π 0.5, QuantVLA attains an average success rate of 97.6%, matching or exceeding the baseline while reducing memory usage from 4.27 GB to 1.28 GB. Similarly, on GR00T N1.5, QuantVLA achieves 88.0% average accuracy with memory reduced from 2.02 GB to 0.91 GB. These results demonstrate that the proposed design effectively mitigates distribution drift caused by quantization in both the language backbone and especially in the DiT action head, which, to our knowledge, has not been quantized in prior VLA work, **thereby delivering state-of-the-art PTQ for VLA models without any retraining.** Additional comparisons with SmoothQuant under different quantization precision settings are provided in Appendix E. Beyond LIBERO, we also further include an extended evaluation on the Simpler [16] manipulation benchmark to assess cross-task robustness. Detailed results are provided in Appendix F.

Efficiency of QuantVLA. QuantVLA achieves substantial memory reduction as shown in Fig. 4. These results confirm that the proposed selective quantization layout

Model	Precision	LIBERO					Memory (GB) (LLM+DiT)	Relative Savings
		Spatial	Object	Goal	Long	Avg.		
$\pi 0.5$	FP16	98.5%	99.0%	97.5%	93.5%	97.1%	4.27	0.0%
+DuQuant(LLM+DiT)	W4A8	86.0%	97.5%	71.5%	50.0%	76.3%	1.17	72.6%
+QuantVLA(LLM)	W4A8	98.5%	99.0%	96.5%	96.5%	97.6%	1.58	63.0%
+QuantVLA	W4A8	98.5%	98.0%	98.0%	96.0%	97.6%	1.28	70.0%
GR00T N1.5	FP16	92.0%	92.0%	86.0%	76.0%	86.5%	2.02	0.0%
+DuQuant(LLM+DiT)	W4A8	66.0%	70.0%	68.0%	76.0%	70.0%	0.74	63.4%
+QuantVLA(LLM)	W4A8	96.0%	94.0%	92.0%	66.0%	87.0%	1.25	38.1%
+QuantVLA	W4A8	96.0%	92.0%	90.0%	74.0%	88.0%	0.91	55.0%

Table 2. Results on LIBERO for different QuantVLA variants on OpenPI $\pi 0.5$ and GR00T N1.5. The table reports success rates (%) across four LIBERO tasks, memory (GB), and the relative memory savings versus each model’s baseline.

and lightweight calibration preserve accuracy while significantly reducing memory consumption. The reduced memory footprint makes QuantVLA particularly suitable for long-horizon policy generation and deployment under tight memory budgets. In practical scenarios, these gains allow the model to process longer temporal contexts, extend input horizons, or run multiple control policies in parallel within the same hardware budget, thereby enabling broader scalability in VLA applications.

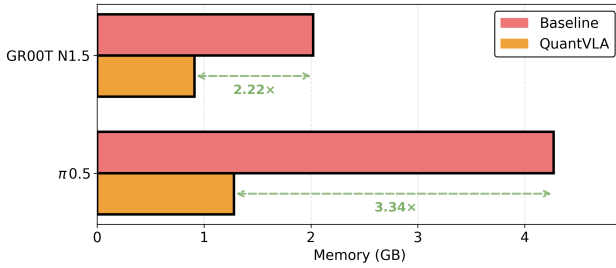


Figure 4. Memory saving of QuantVLA over the baseline on OpenPI $\pi 0.5$ and GR00T N1.5.

Model	Precision	LIBERO				
		Spatial	Object	Goal	Long	Avg.
$\pi 0.5$	FP16	98.5%	99.0%	97.5%	93.5%	97.1%
$\pi 0.5$	W4A8	98.5%	98.0%	98.0%	96.0%	97.6%
+QuantVLA	W4A4	98.5%	98.5%	93.5%	90.5%	95.3%

Table 3. LIBERO results on OpenPI $\pi 0.5$ comparing FP16, W4A8, and W4A4 precision.

Robustness and Generalization Analysis. We further evaluate QuantVLA under two complementary settings to assess robustness and generalization. Table 3 examines the effect of different quantization precisions on OpenPI $\pi 0.5$, comparing the configurations FP16, W4A8, and W4A4.

Model	Denoising Steps	LIBERO				
		Spatial	Object	Goal	Long	Avg.
GR00T N1.5	8	92.0%	92.0%	86.0%	76.0%	86.5%
GR00T N1.5	8	96.0%	92.0%	90.0%	74.0%	88.0%
+QuantVLA	16	96.0%	94.0%	84.0%	80.0%	88.5%

Table 4. LIBERO results under different denoising steps on GR00T N1.5.

The results show that QuantVLA maintains strong performance even at lower bit widths, achieving 95.3% average success rate at W4A4, which demonstrates stable behavior under aggressive quantization. Table 4 evaluates GR00T N1.5 with different denoising steps, where QuantVLA consistently matches or exceeds the baseline, reaching 88.0% average success at 8 steps. These results indicate that QuantVLA preserves task accuracy across precision levels and noise conditions, confirming that the proposed calibration and selective quantization design generalizes well to various inference settings. We further evaluate QuantVLA on OpenVLA in Appendix G, which adopts a non-DiT action head, to assess applicability beyond DiT-based VLA models.

5. Conclusion

We present QuantVLA, the first PTQ framework for VLA models that surpasses full precision baselines without any additional training. With a selective layout, it integerizes the language backbone and the feedforward blocks of the diffusion transformer while attention projections remain in floating point. Two lightweight calibration scalars align attention temperature and restore output energy, thereby stabilizing low-bit inference. As a result, QuantVLA reduces memory and improves accuracy. Overall, QuantVLA is training-free, preserves the original architecture, and is robust across modalities, enabling practical low-bit deployment for long-horizon embodied policies.

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